

# Individual Assessment Coversheet

To be attached to the front of the assessment.

**Campus:** Tyger Valley

**Faculty:** Information Technology

**Module Code:** ITCNA1-12

**Group:** 1

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| Indicate                   | Yes | No |
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| Plagiarism report attached |     |    |

## Declaration:

I declare that this assessment is my own original work except for source material explicitly acknowledged. I also declare that this assessment or any other of my original work related to it has not been previously, or is not being simultaneously, submitted for this or any other course. I am aware of the AI policy and acknowledge that I have not used any AI technology to generate or manipulate data, other than as permitted by the assessment instructions. I also declare that I am aware of the Institution's policy and regulations on honesty in academic work as set out in the Conditions of Enrolment, and of the disciplinary guidelines applicable to breaches of such policy and regulations.

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## Lecturer's Comments:

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# Question 1

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1.1).....A brownout is an example of a power-related issue that occurs when the amount of power that can be provided is less than the amount of power required for a device to function. The symptoms of a brownout are equipment failures including system lockups, productivity loss, data loss and gradual electronic degradation. (Eaton, 2024)

A mechanical failure is a disk-related issue where the spinning plates in a hard drive have experienced physical damage or wear and tear, which can result in permanent data loss. The symptoms of a mechanical failure include abnormal sounds like loud screeching from the hard drive, error messages that are displayed when interacting with a file and files that are missing or cannot be accessed. (Mathe, 2024)

1.2).....To substantiate a theory of probable cause for the power-related issues that the employees are experiencing, I must gather information from the users by asking questions like:

- .....Have you made any recent alterations to your assigned devices, physical or virtual?
- .....Have there been any recent power outages like loadshedding?
- .....Have you encountered anything abnormal while working on the device?
- .....If your device does not turn on at all, do other components like the fan switch on when you supply power?

If the user reports that they have made physical modifications to their device or that other components can receive power, then the theory of probable cause can be that the power cables were not connected properly after the modifications were made.

However, if the user reports instances of power outages and abnormal behavior from their device, then the theory of probable cause can be that the power supply unit or wall socket has been damaged or cannot supply adequate power.

1.3).....To test the theory that the power cables were not connected properly, I can investigate the rear of the computer case to see if the power cable is still attached to the power supply port. If that is not the case, then I can unscrew the side panel of the chassis and determine if all the cables are connected to the correct ports. If one or more cables are attached incorrectly, then my plan of action is to unplug all the cables and, with the use of a guide, connect all the cables into the correct port and test the device to see if it is operational.

Furthermore, to test the theory that the power supply unit has been damaged due to power outages I will first determine if the power supply unit can supply some power by placing one end of a paper clip on the PSU's ground pin and the other pin on the power-on pin as well as switching the power switch on. If the PSU's fan starts to spin, then you I can use a multimeter to determine the amount of voltage that each pin provides (BOHLOOLI, 2023). If the pins do not provide adequate voltage, then it indicates that the PSU is unable to provide enough power to the system. Repairing PSUs is doable; however, it is safer and more cost effective to swap the PSU with a new one (Madsen, 2023).

1.4).....With the use of a word processor like Microsoft Word, I can document the progression of my path to resolving the issues that I have identified. I will document all theories regarding the probable causes for the power related complications with the questions that lead me to those theories. Furthermore, I will extensively document all tests that I conducted and how I did it to determine if my theories were correct as well as all new theories that I had to establish. When I conclude the cause of the issues, I will document, with the time and date indicated, all changes and configurations that I made to resolve the problem and send the whole document to the relevant higher-ups of the business.

## ***Question 2***

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2.1).....The OSI model operates various network communications processes through a protocol-self-determining framework, while the TCP/IP model depend on standardized protocols for communication challenges. The TCP/IP model makes use of four layers instead of seven layers that the OSI model uses. This happens due to the OSI model splitting the functions that the TCP/IP model grouped together into single layers (FORTINET, 2024). The TCP/IP model ensures a connection is present among various computers, while the OSI model assist in standardizing hardware like routers and switches (Williams, 2023).

The TCP/IP model determines how communication can occur by combining both protocols, TCP, and IP, which obtains the packet addressing and forwards or recovers packets.

The OSI model sends the data to other devices by making it travel down the layers and it travels up the layers when receiving the data (CLOUDFLARE, 2024).

The layers in the TCP/IP model include:

- ....The application layer:
  - ... This layer consists of the OSI application, presentation, and session layers.
  - ... It comprises of protocols that operates high-level functions like Dynamic Host Configuration Protocol (DHCP), which provides IP addresses, default gateways, subnet masks as well as addresses of the DNS server to devices.
- ....The transport layer:
  - ... Defines how devices maintain the connections between various application layer protocols.
  - ... Transmission Control Protocol (TCP) can recover lost packets as well as ensures packets that are sent are connection oriented.
  - ... User Datagram Protocol (UDP) forwards packets faster than TCP, because it does not spend time to establish reliable connection resulting in some packets to be lost.
- ....The internet layer:
  - ... Handles packets and combines networks for transmitting the packets (Shacklett, 2021).
  - ... Internet Protocol (IP) offers addressing for packets and channelling through networks.
  - ... Address Resolution Protocol (ARP) gives a host the ability to request MAC addresses related to an IP address.
- ....The link/ network interface layer:
  - ... It is a combination of the OSI data link and physical layers.
  - ... It situates frames onto the physical network and communication occurs on a local network section.

The layers in the OSI model include:

- ....The application layer:
  - ... It allows the user when they wish to transmit data, read messages, or participate in additional network-related tasks to interact with the network.
- ....The presentation layer:
  - ... It translates data for the application layer depending on the criteria that the application accepts (Shacklett, 2021)..
- ....The session layer:

- ... It can create, manage, and terminate any conversations that occurs between multiple applications (Shacklett, 2021)..
- ....The transport layer:
  - ... This layer performs the same functions as the transport layer in the TCP/IP model.
- ....The network layer:
  - ... This layer performs the same functions as the network layer in the TCP/IP model.
- ....The data link layer:
  - ... Manages any issues that are created from bit transmission errors (Shacklett, 2021)..
- ....The physical layer:
  - ... Transfers data through interfaces that are electrical or mechanical (Shacklett, 2021).

The TCP/IP model was described by Vint Cerf and Bob Kahn in 1974, the model was tested and continued to be developed in 1978 and it replaced ARPAnet on January 1, 1983. The OSI model was intended to be a specification of interfaces in 1983 but was formally adopted by ISO as a standard for international use in 1984 (Williams, 2023).

## 2.2).....Synchronous data link control protocol (SDLC):

- ....It is a communication protocol that in most cases is used to manage system network architecture traffic. This protocol acquires all remote devices and connects them, with a point to point or point to multipoint connection format, to a mainframe computer located within a central site. It guarantees that the data units and flow collected from one network point to a different network point is accurate (EDUCBA, 2023). It has the option of being full or half-duplex when transmitting data (IBM, 2023).

## High-level data link control protocol (HDLC):

- ....It is a bit-oriented code transparent synchronous protocol that delivers either connection or connection-less services and obtains different wide-area protocols. Additionally, this protocol is constructed from the SDLC protocol, which allows it to maintain a point to point and multipoint communication format. It supports three modes including:

- ... Normal response mode that transfers the links that are primary to secondary without contention to occur.
- ... Asynchronous response mode that transfers links that are full duplex.
- ... Asynchronous balanced mode, which manages combined terminals that can perform as either primary or secondary (EDUCBA, 2023).

Network control protocol (NCP):

- ....This is a set of protocols that gives users the ability to communicate with multiple devices within a computer network. Moreover, it controls transportation of files and data between the devices. This protocol demonstrates error-free data transmissions by implementing effective data routing as well as flow control. It was important when computer networking was in its early development as it was used in ARPANET, which was a precursor to the internet (DevX, 2024).

Link control protocol (LCP):

- ....It is a standard protocol that is a component of PPP in terms of data communication. It is a vital element of network technology by managing the configuration and testing of the internet protocol connections before any data transmissions can occur. It is designed to create a set of options across devices in a communication link to control the connections in the network. Furthermore, it maintains distinct size limits of packets and can start troubleshooting faulty links that it identifies (DevX, 2023).

Point to point protocol (PPP):

- ....This is the most used protocol in the data link layer, which is essential to capture protocols of higher network layers to pass over communication lines that are either synchronous or asynchronous (GeeksforGeeks, 2023). It outlines the process of generating links when a user receives a connection to the server. In addition, it can determine the authentication across multiple devices, the handshaking as well as how passwords are transmitted.

## Question 3

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3.1).....Network hubs are legacy network devices that transmit messages to all devices located in the network, because hubs do not use MAC or IP addresses to transfer the data. This creates excess traffic on the network, allowing collisions to occur and the network performance to decline. Moreover, hubs are half-duplex meaning that devices connected to the network cannot send and receive data simultaneously further decreasing overall performance of the network.



Network switches are network devices that make use of the MAC addresses of devices to transfer the relevant data over the network. Switches use MAC addresses to determine the destination for the data and transport the data to the correct device instead of sending it to all devices within a network like hubs. This helps to decrease traffic within the network. Furthermore, each port on a switch act as a distinct collision domain, allowing devices to only collide with other devices attached to the same port and further decreases collisions and network traffic. Additionally, switches are also full duplex allowing devices to send and receive data at the same time, which increases the overall performance of data transmissions over the network.

3.2).....Managed switches can work as an unmanaged switch when first used, however it provides the user the ability to configure security settings to protect a network and implement advanced functionality. The security settings include built-in firewall capabilities that protect the network against unauthorized access. Additionally, port speed and duplex mode is also found in the settings, which can be used to improve the overall performance of the network (EXTERNETWORKS, 2024).

However, managed switches have higher downtime and cannot prioritize network traffic (FOCUS, 2024). When downtime occurs, it results in lost data and productivity as users are not able to manage anything on the network (Patrizio, 2024). Furthermore, without the ability to prioritize network traffic, essential services provided by the network will not receive the required bandwidth for it to operate in an effective manner.

3.3).....LANs are installed in a specific location and can span multiple buildings like a campus. WANs are installed in multiple locations and can comprise of multiple interconnected LAN sites.

3.4).....2.4 GHz standard can establish a connection that travels a large area due to its ability to pierce solid materials better than 5GHz but does not support many channels. It is best used in areas like offices or multiple detached buildings. However, in a small building with multiple devices connected to the wireless network, the network will become highly congested and experience interference resulting in less data rates than 5GHz.

5 GHz offers a higher data rate than 2.4GHz, provides more channels and encounters interference less frequently, but it has a smaller coverage area and is not good at going through solid objects. 5 GHz is best used for multiple devices in a small area that does higher bandwidth tasks that takes advantage of its faster speeds and stronger connection.

However, the wider the distance between the devices that use the 5 GHz network will result in the connection becoming worse overall (DAS SYSTEMS, 2023).

3.5).....L1 cache has the lowest latency out of the three cache memories, making it the fastest memory in a computer system, and it holds the data that the CPU requires to complete specific activities. L1 cache has the lowest size as it measures in kilobytes and is entirely dependent on the specs of the CPU (Phillips, 2023). In addition, L1 caches can be found in the processor chip (Lutkevich, 2024). The trade-off of size for speed against size results in fast execution of instructions for the CPU, making the computer system run faster. Future L1 caches can increase in size so that it can hold more instructions for the CPU at a given time, while keeping its rapid speed.

Additionally, L2 caches are secondary caches that are faster than the system memory and are slower than L1 cache, but it is bigger in size as it is measured in megabytes and depends on the CPU (Phillips, 2023). The trade-off of for size against speed can result in making the CPU much stronger the larger this cache becomes. Furthermore, it can be found in embedded on the CPU, coprocessor, or a separate chip (Lutkevich, 2024). Future L2 caches can increase both speed as well as size and make CPUs in general more efficient.

Moreover, L3 caches has the highest latency between the other two caches, making it the slowest cache, but it has the biggest size. Modern CPUs have the L3 cache on itself and the trade-off for biggest size against speed allows it to be treated as a pool of memory that the whole CPU chip can use at any time (Phillips, 2023). Future L3 caches can further increase in size resulting in the performance of the CPU to increase as it can hold more data that can be accessed.

## Question 4

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The following hardware components for the creation of the Windows 10 desktops comprises of:

- .....Intel® Core™ i3-7100 Processor:
  - ... This processor meets the requirements dictated by having 3.90 GHz with two total cores and has a 64-bit instruction set. Additionally, its max size of memory is 64 GB (intel, 2024), which can run tasks that requires a large amount of memory, and this is beneficial for RadioShack. Moreover, it is integrated with the Intel® HD Graphics 630 card (intel, 2024) and this reduces cost as RadioShack is not required to purchase other GPUs.

- ....H110M-A/M.2 motherboard:
  - ... This motherboard is compatible with the processor shown above and has a standard warranty lasting three years (ASUS, 2024). Furthermore, this motherboard provides satisfactory performance, is made of quality components and is inexpensive (Trott, 2017). This motherboard allows RadioShack to purchase quality motherboards in bulk for all their stores at a cheaper price than other motherboards.
- ....Crucial DDR4 2133 MHz 16 GB RAM:
  - ... This exceeds the minimum requirements for RAM by providing 16GB of memory. In addition, DDR4 delivers a high clock speed resulting in faster transfer of instructions between the RAM and processor (Genuine Modules, 2023).
  - ... The 16GB of DRAM ensures users can operate the desktop smoothly when accessing multiple applications simultaneously. This will prove useful when users need to manage the store's accounting and inventory.
- ....Power Supply ATX AK-B1-500 500W:
  - ... A modular PSU that delivers a maximum of 500W of power to the motherboard. This PSU contains several protectors and a PFC filter that helps to prevent interference from the different desktop components (akyga, 2024).
- ....Integral M.2 SATA SSD 120GB:
  - ... It is a solid-state drive, which uses flash memory to permanently store data. It performs the same functions of a hard drive and accomplishes tasks more efficiently compared to mechanical drives.
  - ... The 120GB storage size is great for RadioShack's situation, because it allows it to be low-cost compared to other SSDs. RadioShack can make use of cloud services to store access data anywhere, making hardware storage less necessary.
  - ... This SSD delivers a substantial reduction in loading time and boot ups that the desktop can enact. In addition, it utilizes less power usage, produces less heat, and runs faster than hard disk drives (integral, 2024).
  - ... This form factor resembles a stick of memory that takes less space compared to hard disk drives and 2.5-inch SSDs (Kingston TECHNOLOGY, 2023). This increase space allows for better airflow with the desktop chassis and allows additional SSDs to be installed.

The central processing unit (CPU) is the most important processor in a desktop as it performs program instruction codes sent by software programs and the motherboard ensures that communication can occur between the CPU and other components like the random-access memory (RAM). The compatibility between the CPU and motherboard is important because the change in the design of the CPU between generations makes compatibility with the motherboard only possible if it is the same generation as the CPU. This requires the user to change both the motherboard and CPU to guarantee compatibility when upgrading components.

The Windows 10 Professional edition package is more expensive in comparison to the Windows 10 Home edition package, because it offers additional features not found in the home edition.

Examples of the point above includes:

- ....Professional edition has built-in encryption called BitLocker and has protection for windows information, however encryption in the home edition can only be done through a third-party application (Peters, 2022).
- ....The professional edition allows users to remotely control desktops as well as provides Hyper-V, a virtual machine application. Home edition requires third-party applications to perform these operations (Peters, 2022).
- ....Professional edition provides certain management features like limiting what users of the desktop can do as an administrator, have information available through multiple devices with Microsoft Azure Cloud, acquiring applications and subscriptions in bulk, and easily configure new desktops with dynamic provisioning (Peters, 2022). The home edition does not provide the above features.

The Windows 10 Home edition is best used for individuals due to its cheaper cost; however, the Windows 10 Professional edition is best used for businesses due to the features it provides that assist in wide scale security and management. RadioShack will find the Professional edition the most useful for its over eight thousand stores. All the information that is a part of RadioShack's store accounting and inventory can be stored and retrieved at any time from any device through Microsoft Azure Cloud offered by the professional edition. The use of BitLocker to encrypt and protect information from each device associated with RadioShack adds extra security and helps to prevent cybercriminals from stealing their data.

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